

Lecture 3

Elastically suspended propeller

Part 2 Analysis

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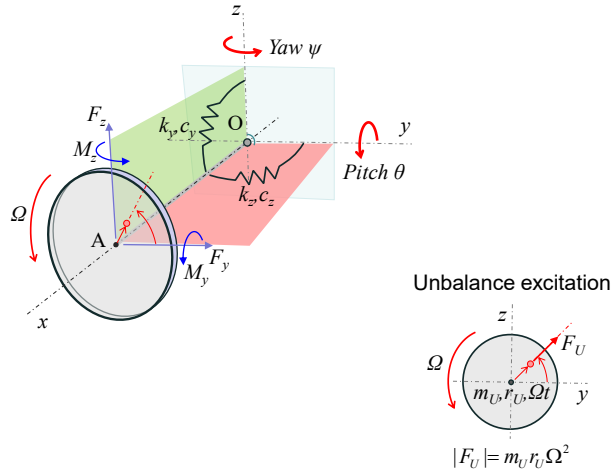
This lecture

- Further analysis in the time and frequency domain
- Reed's model
 - Model property overview
 - Free response and modes of vibration
 - Eigenvalue problem and stability
- 1D and 2D parameter eigenvalue analysis
 - Operational parameters (airspeed)
 - Design parameter (k_y , k_z , ...) stability boundaries
 - Gyroscopic effects
- Example advanced case study

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Baseline Reed's model configuration

All quantities established here follow the "right-hand rule". Notation, frame of reference and sign conventions adopted here are as follows:



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Reed's model equations

Complete form of Reed's model with unbalance excitation:

$$\begin{bmatrix} I_{xx} & 0 \\ 0 & I_{yy} \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} + \begin{bmatrix} c_y & 0 \\ 0 & c_z \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 0 & I_x \Omega \\ -I_x \Omega & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \frac{N_y}{2} K_y R \begin{bmatrix} A_y + a^2 A_x & 0 \\ 0 & A_x + a^2 A_y \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} k_y & 0 \\ 0 & k_z \end{bmatrix} \begin{bmatrix} \theta \\ \psi \end{bmatrix} + \frac{N_z}{2} K_z R \begin{bmatrix} -a A'_x & -A'_y \\ A'_x & -a A'_y \end{bmatrix} \begin{bmatrix} \theta \\ \psi \end{bmatrix} = a R m_U r_U \Omega^2 \begin{bmatrix} -\sin(\Omega t) \\ \cos(\Omega t) \end{bmatrix}$$

Inertia structural damping gyro effect aerodynamic damping structural stiffness aerodynamic stiffness unbalance loading

Compact vector-matrix form of Reed's model:

$$\mathbf{M} \begin{bmatrix} \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} + (\mathbf{C}_S + \mathbf{C}_{gyro} + \mathbf{C}_{aer}) \begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + (\mathbf{K}_S + \mathbf{K}_{aer}) \begin{bmatrix} \theta \\ \psi \end{bmatrix} = \mathbf{f}(t; \Omega)$$

Reed's model – 2nd order system of two coupled ODEs with nonzero RHS:

$$\mathbf{M} \ddot{\mathbf{q}}(t) + \mathbf{C} \dot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) = \mathbf{f}(t; \Omega), \quad \mathbf{q}(t) = [\theta(t), \psi(t)] = ?$$

Methods of solution and analysis:

- Free vibrations – eigenvalue analysis, initial conditions, ...
- Forced vibrations – harmonic (e.g., unbalance) loading, transients, ...
- Advanced topics – nonlinear problems and stability, model refinements, ...

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Free vibrations and eigenvalue problem

Free aeroelastic response (zero RHS):

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{D}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$$

Assumed solution:

$$\mathbf{q} = \mathbf{Q}e^{st}, s \in \mathbb{C}$$

$\mathbf{Q}$ is a complex-valued vector of amplitudes
 s is a complex number.

Substitute assumed solution to obtain **quadratic eigenvalue problem**:

$$(s_j^2 \mathbf{M} + s_j \mathbf{D} + \mathbf{K}) \mathbf{Q}_j = \mathbf{0} \quad s_j, \mathbf{Q}_j = ? \quad j = 1, 2, \dots, 2N$$

$$2N = 2N_C + N_R$$

N ... number of DOFs
 N_C ... number of complex eigenvalue pairs
 N_R ... number of real eigenvalues

Methods of solution:

- *Direct method* by solving the characteristic equation (see below)
 - Good for analytical solution tasks and theoretical insights.
 - Use "roots" function in Matlab.
- *Indirect method* by solving an equivalent 1st order ODE problem
 - Good for practical solution tasks in numerical setting.
 - Use "eig" in Matlab

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Eigenvalue problem – direct method

Nontrivial solutions can be found by solving the characteristic equation:

$$\det(s^2 \mathbf{M} + s \mathbf{D} + \mathbf{K}) = 0$$

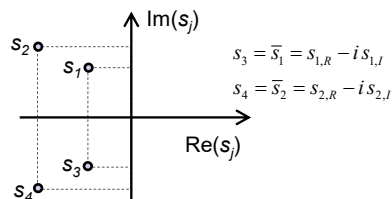
This is a quartic polynomial with 4 complex-valued roots (eigenvalues):

$$a_1 s^4 + a_2 s^3 + a_3 s^2 + a_4 s + a_5 = 0$$

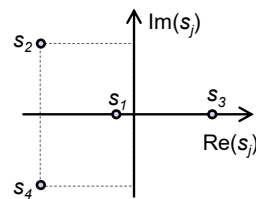
$$s_j, \quad j = 1, 2, 3, 4$$

All complex-valued eigenvalues appear as complex conjugate pairs.

Examples of eigenvalue solutions for 2DOF system:



Example 1: all complex eigenvalues



Example 2: mixed real and complex eigenvalues

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Eigenvalue problem – indirect (numerical) method

To solve the original 2nd order ODEs numerically, we express the equations in the 1st order ODE form. Transformation from the *physical space* to the *state space* realises this operation. 1st order ODE systems are solvable using generic numerical integration (time stepping) and eigenvalue solvers in various computational environments (Matlab, Python, etc.)

General EOM rearranged:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{f}(t) \quad \text{2nd order ODEs}$$

$$\mathbf{M}\ddot{\mathbf{q}} = -\mathbf{C}\dot{\mathbf{q}} - \mathbf{K}\mathbf{q} + \mathbf{f}(t)$$

$$\ddot{\mathbf{q}} = -\mathbf{M}^{-1}\mathbf{C}\dot{\mathbf{q}} - \mathbf{M}^{-1}\mathbf{K}\mathbf{q} + \mathbf{M}^{-1}\mathbf{f}(t)$$

$$\ddot{\mathbf{q}} = \begin{bmatrix} -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix} + \mathbf{M}^{-1}\mathbf{f}(t)$$

Mathematical identity to augment the EOM:

$$\dot{\mathbf{q}} = \dot{\mathbf{q}}$$

$$\dot{\mathbf{q}} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix} + \mathbf{0}$$

Combine the two systems of equations:

$$\begin{bmatrix} \dot{\mathbf{q}} \\ \ddot{\mathbf{q}} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{f}(t) \end{bmatrix}$$

$$(s_j^2 \mathbf{M} + s_j \mathbf{D} + \mathbf{K}) \mathbf{Q}_j = \mathbf{0}$$

Two alternative formulations of the same eigenvalue problem

$$(\mathbf{A} + s_j \mathbf{I}) \mathbf{X}_j = \mathbf{0}$$

$$\mathbf{X}_j^T = [\mathbf{Q}_j^T, s_j \mathbf{Q}_j^T]$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t) \quad \text{1st order ODEs}$$

$$\mathbf{x} = \begin{bmatrix} \mathbf{q} \\ \dot{\mathbf{q}} \end{bmatrix}, \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix}, \mathbf{g}(t) = \begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{f}(t) \end{bmatrix}$$

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Free vibration response

The total free vibration response can be written as (four combined partial solutions):

$$\mathbf{q}(t) = A_1 \mathbf{Q}_1 e^{s_1 t} + A_2 \mathbf{Q}_2 e^{s_2 t} + A_3 \mathbf{Q}_3 e^{s_3 t} + A_4 \mathbf{Q}_4 e^{s_4 t}$$

Scalar constants A_i depend on four initial conditions (initial pitch, yaw and their rates of change).

Each solution component influences the nature of the total solution as follows:

$$\mathbf{Q}_j e^{s_j t} = \mathbf{Q}_j e^{(s_{j,R} + i s_{j,I}) t} = \underbrace{\mathbf{Q}_j}_{\text{Eigenvector}} \underbrace{e^{s_{j,R} t}}_{\text{Exponential decay or rise}} \underbrace{(\cos(s_{j,I} t) + i \sin(s_{j,I} t))}_{\text{Harmonic oscillation at } s_{j,I}}$$

$$\mathbf{Q}_j = \begin{bmatrix} \theta_{j,0} \\ \psi_{j,0} \end{bmatrix} \in \mathbb{C}^2$$

Complex angular amplitudes

Hence, to understand the system's dynamics, we need to study all its modes:

- Stability: eigenvalues (damping information)
- Mobility: eigenvectors

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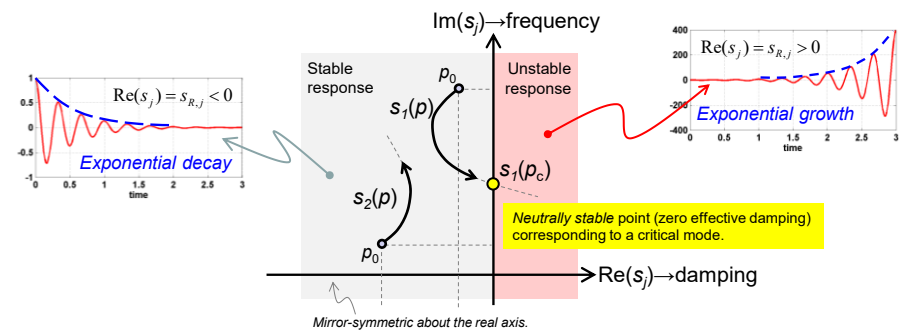
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Stability

Eigenvalues: As the eigenvalues are complex-valued quantities, where each contains information about the frequency and evolution of the modal participation (decay or rise), they are frequently presented in the complex plane. The eigenvalues constitute a set of points in the *complex plane*.

Stability: The distribution of these points is indicative of the system's *stability*. The system is stable if all its eigenvalues reside in the *left half-plane*, meaning that they all have the negative real parts.

Evolution: The eigenvalues depend on the (model) parameters. If one parameter is varied, each eigenvalue changes (typically) smoothly along a curved trajectory called the *modal locus*. The penetration of the locus through the vertical (imaginary) axis indicates a point of neutral stability followed by the *loss of stability*. This intersection point corresponds to the *critical value of the varied parameter*. The nature of the penetration suggests a *type of instability*.



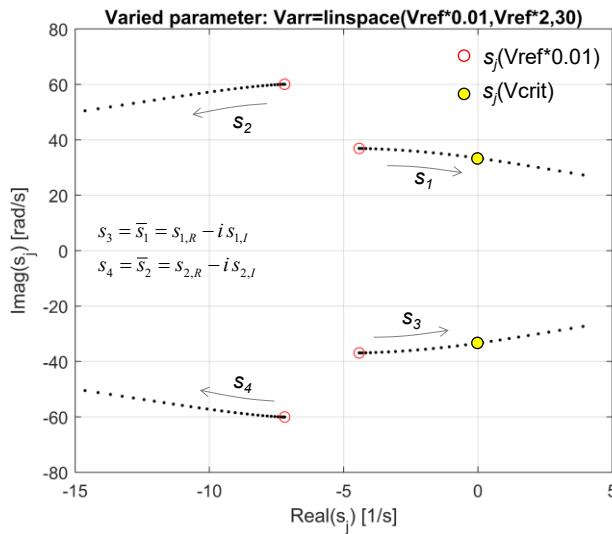
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Reed's model stability – 1D parameter study

Consider eigenvalue analysis performed on Reed's model (L2) while varying the *freestream velocity* parameter and plot the *eigenvalue loci*.



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Annotated 2D eigenvalue stability analysis in Matlab

Copy and paste to Matlab editor, save as "batch_reed_2d.m" and then execute to study the whirl dynamics.

Problem parameters

Parameter ranges

Reed's model

Eigenvalue solver

Visualisation

```
function batch_reed_2d
% Parameters
R=0.152; % m, Rotor radius
Omega=40; % rad/s (-1), Rotor angular velocity
a=0.25; % -, Pivot length to rotor radius ratio
Ix=0.000103; % kg*m^2, Rotor moment of inertia
Iy=0.000178; % kg*m^2, Macell moment of inertia
Cdt=0.001; % N*s/m^2, Structural pitch damping
Ktht=0.4; % N*s/m^2, Structural pitch stiffness
Cpsi=0.001; % N*s/m^2, Structural yaw damping
Kpsi=0.4; % N*s/m^2, Structural yaw stiffness
Nb=4; % -, Number of blades
c=0.026; % m, Blade chord
claw=2*pi; % rad (-1), Blade lift slope
rho=1.225; % kg/m^3, Air density
m_u=0.001; % kg, unbalance mass
r_u=0.001; % m, unbalance radius/offset
V=1.37*Omega*R; % m/s (-1), Freestream velocity (nominal V=6.71)

% Parameter range and loop
Np1=30; Np2=30;
p1=linspace(Ktht*0.01,Ktht*2,Np1); p2=linspace(Kpsi*0.01,Kpsi*2,Np2);
SS=zeros(Np1,Np2);

for ii=1:Np1
    for jj=1:Np2
        Ktht=p1(ii); Kpsi=p2(jj); % Vary the chosen parameters

        % Aerodynamic constants and integrals
        Kal=0.5*rho*c*V^4*Omega^2; COEF=(Nb/2)*Kal*R; mu=V/(Omega*R);
        A1=(c/R)*quad(@(etal) (mu.^2./sqrt(mu.^2+etal.^2)),0,1); Apl=mu*A1;
        Ap2=(c/R)*quad(@(etal) (mu.^2.*etal.^2./sqrt(mu.^2+etal.^2)),0,1);
        A3=(c/R)*quad(@(etal) (etal.^4./sqrt(mu.^2+etal.^2)),0,1);
        % Matrices
        M=diag([Iy,Ix]); % Mass
        C=diag([Cdt,Cpsi]); % Structural damping
        Cg=Omega*[0,Ix;-Ix,0]; % Gyroscopic
        K=diag([Ktht,Kpsi]); % Stiffness
        Kae=COEF*[-a*Apl,-Ap2;-a*Apl]; % Aerodynamic stiffness
        Caer=(COEF/Omega)*[A3*a^2*A1,0;0,A3*a^2*A1]; % Aerodynamic damping
        % Assembled Reed's model system and state space matrix
        NBNb; CC=Caer+Kae; KBN=Kae;

        % Eigenvalue analysis
        s=eig(ss+(ss(2)+eye(2))-inv(NBN)*KBN,-inv(NBN)*CC);
        SS(ii,jj)=max(real(s));
    end
end

% plotting
figure, surf(p1,p2,SS), hold on
contour3(p1,p2,SS,[0,0],0,'r','linewidth',2)
plot3(p1(1,end,end,1,1),p2(1,1,end,end,1,1),[0,0,0,0,0],'-m')
xlabel('ktheta [e.u.]'), ylabel('kpsi [e.u.]'), zlabel('max(real(s)) [1/s]')
view([110,40]), colormap(pink)
```

Explore the influence of the support system parameters

2D (2-loop) variation of two chosen parameters (support stiffness)

Store critical real parts of the eigenvalues

Zero level plane cut through the critical eigenvalue surface (neutral stability)

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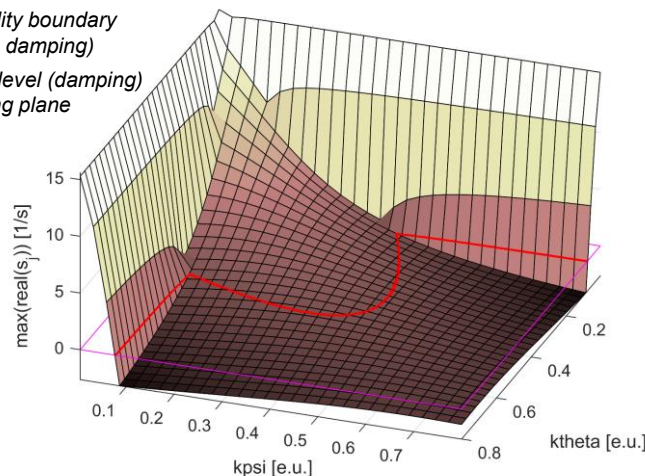
Reed's model stability – 2D parameter study

Consider the eigenvalue analysis performed on Reed's model (L2) while varying *pitch* and *yaw stiffness* parameters and plot the stability boundary.

Varied parameters:
 $p1=linspace(Ktht*0.01,Ktht*2,Np1); p2=linspace(Kpsi*0.01,Kpsi*2,Np2);$

— stability boundary
 (zero damping)

— zero level (damping)
 cutting plane



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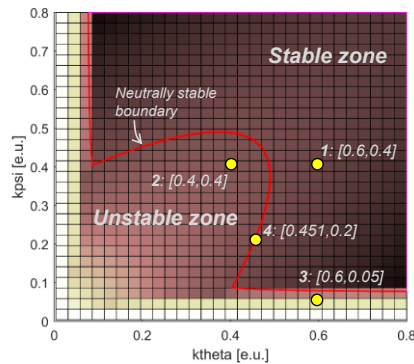
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Reed's model stability – 2D parameter study

Consider the eigenvalue analysis performed on Reed's model (L2) while varying *pitch and yaw stiffness parameters* and plot the stability boundary.

Top view of the stability zone:
The pitch-yaw stiffness stability diagram



To investigate these regimes closer (e.g., the nature of the shaft's whirl), execute the Reed's model implementation (L15, s.13) with appropriate parameters and study the resulting propeller hub orbits.

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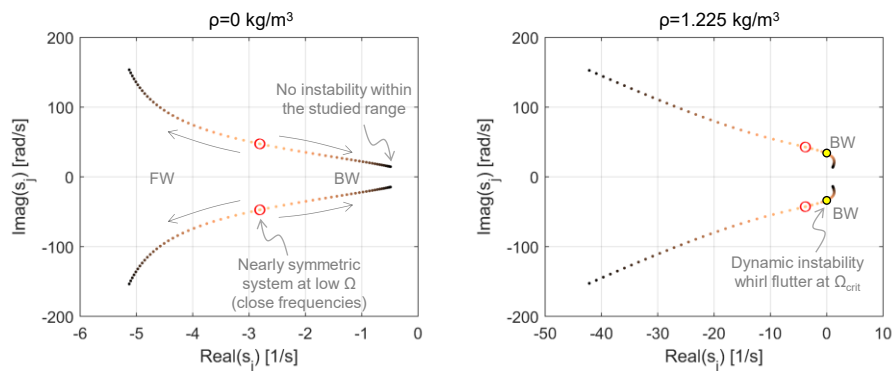
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Reed's model stability – gyroscopic effects

Consider the eigenvalue analysis performed on Reed's model (L2) while varying rotor speed, in air and vacuum conditions, and plot the eigenvalue loci.

Varied parameter: $\Omega_{\text{garr}} = \text{linspace}(\Omega_{\text{garr}} * 0.01, \Omega_{\text{garr}} * 6, 30)$



Note gradual splitting of originally identical (two pairs) of natural frequencies. One pair corresponds to the *backward whirl* (BW) and the other pair to the *forward whirl* (FW). In this (model) context, BW is responsible for the loss of stability. This may not be true for more complex wing-nacelle-propeller whirling configurations. Higher fidelity models are necessary (see the next slide).

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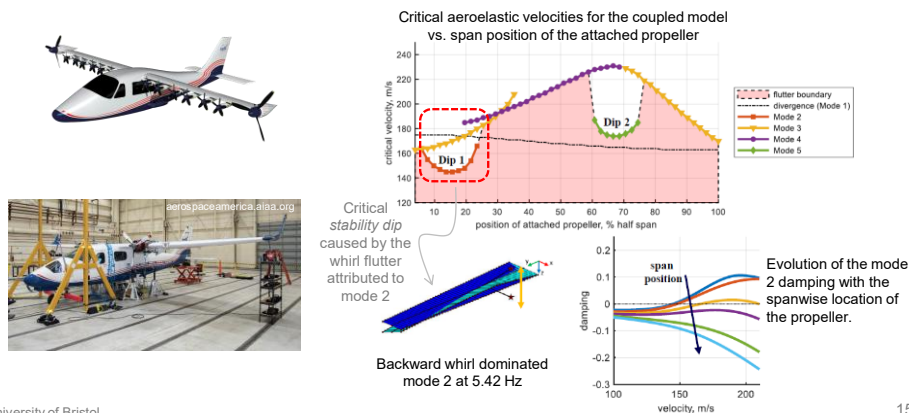
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Example – NASA electric flutter

Consider an example of all electric fixed-wing aircraft with distributed propellers across the wing-span (broadly inspired by NASA Maxwell X-57 demonstrator).

Böhnisch, N., Braun, C., Koschel, S. and Marzocca, P., 2022. Whirl flutter for distributed propulsion systems on a flexible wing. In AIAA Scitech 2022 Forum (p. 1755).

A chosen example demonstrates the impact of the propeller placement variation on the overall aeroelastic stability of the wing. In particular, this example shows the interaction between the classical wing flutter modes and the propeller-nacelle dominated modes. It also evidences the critical impact of the propeller-driven instability on the overall design.



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Summary

- Observations from eigenvalue analysis:
 - Presence of damping and gyroscopic matrices leads to a quadratic eigenvalue problem – solved directly (using characteristic equation) or indirectly (via an equivalent 1st order ODE problem).
 - Complex-valued eigenvalues possess certain structure (complex/real, conjugacy) and determine the free response of the original problem.
- Analysis of stability:
 - Real eigenvalue parts determined the decay/rise of the modal participation – used to determine the system's stability. To ensure stability, all eigenvalues must have negative real parts (decay).
 - Different types of instability (static/dynamic) can be exercised by Reed's model when it reaches the critical values of its parameters.
 - Stability boundaries and critical values (including associated safety margins) are computed to design the system or prescribe its operation.
- Practical implications:
 - Practical applications call for refined model which include both propeller and wing dynamics – propeller dynamic contribution can constitute the main performance or design driver.

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